

VALENTIN VELEV

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Frankfurt, Germany

RESEARCH INTERESTS

My research interests lie in developing accessible language models and language model-based systems for multilingual and context-aware real-world applications. I am particularly interested in exploring how such models can be used for social good, including supporting education, advancing climate science, and improving medical information access and communication.

EDUCATION

MSc Data Science (SEDS)

University of Konstanz

October 2023 – March 2026

Konstanz, Germany

- Preliminary average grade: 1.1
- Courses: Deep Learning (1.0), Word Representations and Language Models (1.0), Statistical Learning (1.3), Data Mining (1.0)
- MSc thesis on the evaluation of document retrieval models submitted to CIKM 2026

BA Political Science

Goethe University Frankfurt

October 2019 – January 2023

Frankfurt, Germany

- Final grade: 1.2 (graduated with distinction)
- Courses: Statistics (1.3), Data Science (1.0)

EXPERIENCE

Data Scientist

Codify AG

March 2025 – Present

Kreuzlingen, Switzerland

- Designed and implemented evaluation pipelines for RAG and agentic AI systems using Python, MCP servers (for agent tools), LLM-as-a-judge metrics, and Arize Phoenix (for agent traces), improving the reliability of model assessment and enabling scalable benchmarking across multiple use cases
- Built an LLM conversation evaluation framework for multi-turn interactions to assess model performance
- Integrated end-to-end evaluation workflows and visualizations into production systems using Python, TypeScript, and SQL
- Designed prompt injection challenges for workshop at the AIXCHANGE Conference 2025

Teaching Assistant

Department of Computer and Information Science, University of Konstanz

August 2025 – February 2026

Konstanz, Germany

- Provided technical guidance for the master's-level course "Word Representations and Language Models," covering embedding models, sequence-to-sequence models, the Transformer architecture (including BERT), and generative language models
- Developed and maintained reproducible teaching materials using Python, Git, and $\LaTeX$

Student Research Assistant

Center for Data and Methods, University of Konstanz

April 2024 – February 2025

Konstanz, Germany

- Conducted independent quantitative analyses and contributed to three peer-reviewed journal publications
- Designed data visualizations and reproducible analysis pipelines in R and Python for publication-quality outputs
- Collaborated on research design, data preprocessing, and statistical modeling using R, Python, Stata, and $\LaTeX$

PUBLICATIONS

Velev, V., Fuchs, J., & Spitz, A. (2026). An R Package for Visualizing Significance in Directed Pairwise Relationships Using Tapered Edge Glyphs. *Proceedings of the 28th Eurographics Conference on Visualization (EuroVis 2026) – Posters and Demos*. <https://doi.org/10.2312/evpd.20261030>

Velev, V., & Schmidt-Catran, A. W. (2024). Economic Inequality and Public Support for Redistribution in Europe: A Cross-Sectional and Longitudinal Multilevel Analysis. *International Journal of Public Opinion Research*, 36(4), edae055. <https://doi.org/10.1093/ijpor/edae055>

Velev, V. (2023). Replication of "Multilevel Models for the Analysis of Comparative Survey Data Common Problems and Some Solutions" with R. *SocArXiv*. <https://doi.org/10.31235/osf.io/phq56>

SKILLS

Technical:

Machine Learning & NLP: Transformers, PyTorch, RAG, LangChain, LlamaIndex, supervised and unsupervised LLM evaluation, agentic AI, MCP servers, fine-tuning pipelines, information retrieval

Experimentation: Experimental design, error analysis, benchmarking, reproducible research

Programming: Python, SQL, R, TypeScript, Java, HTML/CSS

Systems & Tools: Docker, Git, APIs, L^AT_EX

Data Analysis & Visualization: pandas, matplotlib, Stata, Tableau, Data Studio

Languages: English (C2), German (native), Bulgarian (native), Dutch (elementary proficiency)